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REINFORCEMENT LEARNING

Function Approximation

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So far we have been talking about mostly value function based approaches for solving RL problems right, so whether it is Monte Carlo or TD or whatever right so it has been value function based and there is a very small fraction that we looked at we looked at policy gradient methods and there was a reinforce algorithm we looked at, but we never talked about how would you use policy gradient for solving the full, full RL problem right we only talked about it in the context of the immediate RL, right.

So I will come back to that at some point I will talk about policy gradient methods for the full RL problem right, and perhaps before that I will do a little bit of average reward reinforcement learning before we get onto that, right. But except for the policy gradient part right, and the linear bandits right, except in those two circumstances we have essentially been using some kind of a,

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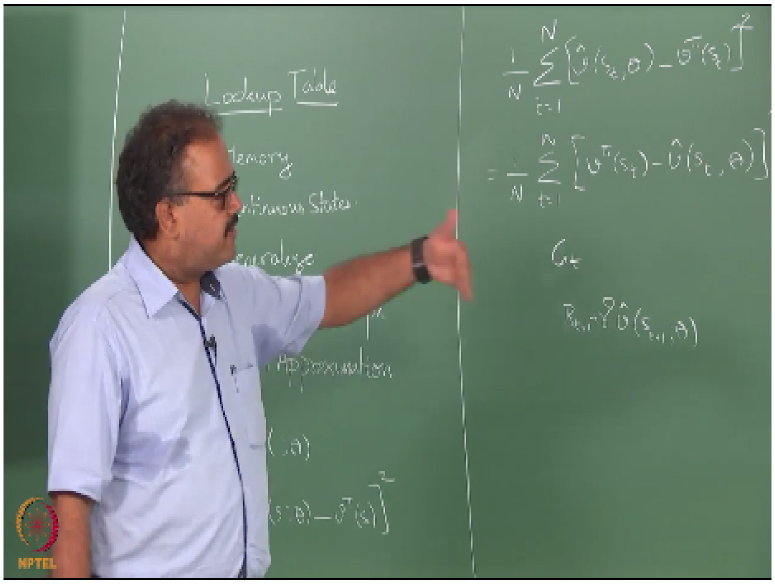

Some kind of a lookup table representation for our value functions right, whether it is a bandit case where we kept a Q of a right, or when we talk about V(s) or V of, I mean Q (s,a), right we have always been assuming some kind of a tabular representation right, for every s I know what the value is right. So I have been looking at some kind of a tabular representation so what are the problems with this tabular representation.

So what are the drawbacks of using a lookup table, are there any drawbacks for using a lookup table, memory right. Yeah, good. For the time being let us assume that, we will come to that. So, first question is memory, okay next problem is handling continuous states right, anything else. You are not allowed for the remainder of this class you are not allowed to answer questions, if people are wondering, we just had this discussion yesterday in the lab meeting so in fact he is using a cheat sheet.

Let us suppose that memory is not an issue right, so we have huge RAM machines you know I mean so 1 terabyte RAM is not uncommon now, right let us say that I build a very, very large memory machine right, or I use some external memory algorithm for something all you need is some kind of a large table data structure, you need to have a data structure that can handle very large tables and in fact there actually exists external table storage libraries right, that allow you to handle tables as large as your hard drive, right.

So let us say memory is not an issue right, and let us say continuous states is also not an issue, I am leading up to something else, right so I have discrete states right, and I have a lot of them fine but I also have lot of memory that I can store these things right, so what else could be an issue in having a lookup table representation. What is it, query time could be high okay, yeah, something non-implementational, maybe I should not say it is non-implementational but something non system level answer.

Guys who know machine learning who have been in the ML class and things like that should have an answer for this even if not others, no, come on okay, so what are we talking about here we are talking about functions right, we are talking about value functions right, we are talking about state value functions and action value functions, v and q. So we have learnt about functions right, in machine learning we have looked at functions right, so have we ever used a lookup table for learning a function from data, what do we do.

Use some parameters and try to estimate those parameters, so why did we do that, so how does learning work right I give you a set of data points right, and from those examples you are supposed to induce forms of functions that allow you to answer questions about points you have not seen already, right. So why does it become important in RL if I have lots and lots of states I have a huge table and if I am using the classical lookup table version of things then unless I have visited that state,

Not once, unless I have visited that state often enough, I do not have an estimate for that value function, right that means if I have like a million states or million is small right, so in fact there is one problem which I implemented some algorithms on during my PhD., which had 10^{25} states right, and if I have to experience every state in that multiple times, I would still not have graduated, right.

So we need something else right, so we need to be able to generalize. I feel like a dentist, extracting answers is like extracting a tooth from you guys I mean anyway fine, let us go back so we want to generalize right, that is very crucial so to achieve all of these what do we do, instead of representing the value function as a lookup table we try to represent it as some kind of parameterized function, right. So the most popular choice for the last couple of years has been, has been to use a neural network, right use a neural network, an artificial neural network as your parameterized representation.

So what are the parameters here, they will be the weights of the neural network right, so speaking of which so pretty momentous thing happened a few minutes ago so how many of you did you follow the α go thing, it won yeah, so α go won against the highest ever rated human master in the game go, so remember I told you about α go which is a RL and deep learning based go playing agent just so, the matches have started in Seoul, right so this guy is a Korean who is the highest ranked ever human in go and the first game α go won.

And people are actually stunned even the deep mind guys did not expect to win, right the company who made this thing they just wanted to see how close they can run the human right and they won maybe he was sick or something I do not know we will see so it is a five match series million dollar prize money they probably win it and then donate it to some charity or something, but anyway so that is amazing right.

So and that uses a neural network for representing the value function okay, so the parameters we talked about here are your the weights of the neural network right, it could be any kind of weights, thresholds right and other parameters slopes of the transfer function whatever right, however complicated you want to make the parameterization right, so that is typically what we do, right. So this kind of parameterized representation obviously, comes with a cost, so what is the cost, what is it, there will be some kind of underlying bias and what does the bias lead to and it leads to some kind of approximation right,

So as soon as I move into this kind of a parameterized representation unless I make the parameters parameterization so powerful that any value function that I want to represent I am able to put it in here which will be kind of defeating the purpose right, I will be essentially end up with as large a state space as I originally had right as I mean so if you think about it, the look up table itself is a parameterized representation, what are your parameters here each entry a value of each state itself is a parameter, right and you can just think of that as a parameterized representation right.

So if I make my parameterized representation complex enough that it can represent any value function without any loss right, then the number of parameters I will need will essentially be the same number of entries as the lookup table. So I will not have saved much right, so you essentially you want a simpler representation you do not want something that has as many parameters as the lookup table you want something that has a lot fewer parameters than the number of states and perforce you have to accept an approximation, right.

Because I will not be able to model every value function defined on some state so i will have to essentially accept some kind of an approximation, so this is also known as, right, so people call this function approximation, RL and function approximation and so on so forth, the chapter itself is called, in the new book I forget what, the on policy evaluation with function approximation or something like that so that is the title of the chapter right.

Unfortunately since the chapter is not edited it also talks about control in the same chapter and the second chapter is called on policy control with function approximation and it is empty the tenth chapter is called on policy control with function approximation is empty and the on policy evaluation with function approximation talks about both right now, correct, okay so it is called function approximation essentially you can use this as synonyms, right.

You can say you I am using a parameterized representation or function approximation typically are used synonymously ok great. So what is going to happen so I have my true value function say $v\ \pi$ right, so I am going to approximate that by some function v hat, which is defined by a set of parameters θ, so we looked at this notation earlier right, in reinforce when you looked at reinforce I looked at this kind of a notation, this when I put a semicolon and write something to the right of the semicolon it is the set of parameters to the left of the semicolon is the argument of the function, ok great.

So what should be the criteria for deciding on what $\hat{v}$ should be, so there are multiple things I need to do here right, so for the benefit of people who have not done ml right there are at least two stages that I need to proceed here. The first stage is deciding, hmm, that is the second stage, the first stage is deciding how the output is going to depend on θ, I am saying that some set of parameters, right there must be some functional form associated with these set of parameters and I said neural networks, that is one functional form, right.

So, parameters are the weights right, I can very simply write $\hat{v}$(s); w, but how do those w determine what v(s) is, right, so that is some kind of a connected neural network I can talk about three layers, four layers or 150 layers I mean if you work for Google or Microsoft you can start talking about 130 layers of a neural network and things like that right, so how do I this is called deciding the functional class right, or deciding on the class of models that you are going to use for representing this, right is a first thing.

What is it that I am going to do I can use decision trees right I can use regression trees in fact people have used regression trees for representing value functions right, so that is one option right I can use some kind of complex aggregation, state aggregation I will talk about that little later because that is a very popularly used function approximator, right and we can use neural networks another very popular function approximator is just to use some kind of a linear parameterization.

So what I mean by that so I will have θ 1, θ 2, θ 3 and then it will just be the weight and I will assume that my state is represented by some features, right if it is a grid world it can be the x coordinate and the y coordinate right, or if it is some kind of robot it can be the x coordinate the y coordinate the angle the velocity with which the bot is moving right and the angular velocity and all of those things x velocity y velocity there could be a vector of features right, let us say that my state is represented by some vector right phi, so φ1 of s will be the x coordinate, φ2 of s will be the y coordinate and so on so forth, and my value of a state will be φtranspose θ right, θ 1

$\varphi1 + \theta\ 2\ \varphi2 + \theta\ 3\ \varphi3$ and so on so forth that will be my value function this is called a linear parameterization, right.

So what we will do later right is explore linear parameterization in more detail but this is a very popular kind of parameterization I can basically the first thing I have to do is decide on what is the class of function approximation I am going to look at, right and what is the second thing I have to do, so what is the second thing I have to do right, now I need a error measure right, to figure out what is the performance measure I am going to use right and in some sense this can be common across different choices of parameterization right.

It need not depend on the choice of parameterization what error measure I use need not depend on the choice of parameterization but what would potentially depend on the choice of parameterization is how do I find that θ that minimizes this error measure, okay. So there are three things now right, so the first thing is find out the family of functions that I am considering whether it is a neural network or decision tree or whatever.

The second thing is well figure out what is the error measure how I am going to decide that I have a good approximation to the function and how am I going to find that θ how I am going to find the θ that gives me this good approximation, okay so that is basically the setup for us these are the questions that we will have to ask, right and so what do you think is a good measure, least squares, least squares is good enough?.

Well it is regression what else you want, right so least squares is pretty good enough right so how will least squares work, least squares is the target that I want, the performance measure itself would be mean squared error, so essentially what I will be looking at is, if it is continuous I am going to have a integral there, is this fine, yeah so I left a significant gap here so that should mean something other than 1 by N within the summation, right, is 1 by N fine.

So what is the problem with 1 by N so we said mean square right, so mean so you have to take the mean of the square the square of the errors obviously right, so if I do it by 1 by N, what does it mean, that is a different mean yeah, so what does it mean, I am weighing each state uniformly I want to get a good approximation everywhere right that is what I am trying to do right when I do 1 by N, I am trying to get a good approximation everywhere right. If you remember the expected error computation that we did in, how many of you were in ML, good number right okay yeah.

If you remember the expected error computation in ML, this you have to cast your mind back to like the third class or fourth class in ML right, how did we write down the expected error computation there, weighted by the probability of that particular pair occurring right so likewise and then what we what we said we assume that the data is coming to us actually from the probability distribution and therefore we can do the empirical mean you can just divide by 1 by N

Because if our data point is more likely we will assume that it appears more often in the sample that is given to you right that is how we ended up justifying using 1 by N right, likewise what would be the probability here. So we are lucky in one way because we have a π here right, and therefore I am going to write some new quantity here any guesses what that is, how many times no, close, elec guys there is a technical term for it.

Yeah that is what it is but it is called the, let us step back again, I have an MDP I fix a policy π right, and then I am looking at the evolution of the system what is it going to be, mathematically it will be a Markov chain okay, it is no longer an MDP right so what do you think d π of s is, the steady state distribution under the policy π right. So it is essentially it becomes a Markov chain, if I keep running the Markov chain for a long time right, very, very, very long time it is going to settle into some visitation patterns right.

So the steady state distribution essentially tells me if I, steady state probabilities essentially tells me if as T tends to infinity I sample a state, I stop the Markov chain running and I sample a state where it stopped at right, the probability that I will find it in that state that is essentially what d π s is. Well I do not have any decisions to make any more right, my π is fixed so when I come to a state I just toss a coin pick the action according to the π and I go on right, my π is not changing so if I look at the overall transition dynamics right it will just look like, it will just look like a Markov chain, right. In fact we did this when we did that convergence proofs right for like

Bellman equation so we defined the transition matrix itself I took the out action out if you remember I wrote p π of j, i, j, s, so when we did that we took out the effect of the policy there, we essentially wrote the Markov chain dynamics back there, so essentially that is what we are doing here again so d π of s is the steady state probabilities of state s under the policy π.

So this essentially tells me that the states that I am, I am more likely to visit under this policy if I keep running this policy often those are the states for which I want the approximation to be good, okay. Now let us go back to our sampling case, if I am trying to instead of knowing this d π s, if I am trying to solve this using sampling, right so what I am I asking for here what kind of sampling I am asking for here, you are allowed to answer this question on policy, right.

Because I expect on policy samples I do not want you to give me arbitrary states it has to be sampled according to the policy π because only then it will approximate the d π s okay, so I am just building up when we have talked about all of these things individually but you can see that it all hangs together, even when I am doing function approximation right, when I am doing this, this is the thing I want to optimize so I can do it like this, take that as my error measure right provided these were are all sampled, s 1 to s N were sampled along a trajectory generated by using pi, okay makes sense, good.

So now we have an optimization, we have a least squares set up right, I mean I have an objective function so can I go and optimize that and find θ, I mean let us forget about what is the functional form of θ now is, there are still some questions I can ask in the general setting so can I just take this and go optimize the θ, other way I should write it slightly differently so that it helps me later, okay there you go, so this makes it easier right, so this is target right, minus my current estimate so this is the error, right so target minus current estimate is the error, so i take the square and so, instead of I mean writing it like this does not fit with our target minus estimate formulation that we had earlier so I just flipped it around.

Since the squared error it is okay, right so is there any problem in trying to optimize this, there is a problem, there is a very, very serious problem what is the problem. Even before that even before we talk about solution techniques, exactly, I do not know what v π of s is, if I know v π of s I have no problem I do not even have to learn anything right, I put v π as my target and I do not have the target what will I learn about, correct okay.

So we have to find out a way of specifying the target right, so what should be a good target, we spoke about this, the same thing we did in the tabular case so G t will be, what, that will give you Monte Carlo right, so what else can you do, you are using, where, no, we do not know v π that is what I told you right we do not know v pi, see classically if for regression problems you would assume that you have data of the following form some x and F of x, x F of x, x F of x, and then you go and learn a F hat x correct, so that is the normal assumption that you make.

But here the training data will not be of that form the training data is going to be of what form, will be trajectories the training data will be trajectories, I will have S t, A t, R t + 1, S t + 1, A t + 1, R t + 2, S t + 2 A t + 2 R t + 3 and so on so forth that is how my training data is going to look like nowhere I am going to tell you okay, here is S t, here is v π of S t right, so this is the difference so in classically in regression the training data will be x, F x pairs and from here you will learn some kind of F hat x right.

Here for RL the training data will be right, so somehow from this we need to learn $\hat{v}$ correct, so the training data should start looking something like this right, from here I can create, here is one set of training data I will create right, whatever is the state S t right comma, that is the first data point, second data point will be like that, right so i will create data points like this and I can use it for training right, so alternatively what can I do right, so this is how I will create training data for my, if I am going to use my regression as a black box right.

My training data for the regression will look like this right, so this seems to be a little acceptable even though there are problems here but this seems to be a little acceptable than that why, what is the problem with that, so what do you do, you go back and do it again right, that is where you get into the problem because what you are now setting up is a non-stationary target, so i will make up a set of training data like this right, or let us say I will do one training data like this right and then I go back I will update the θs right, so i will get a new estimate for my parameters θ and then I come back and try to do the same sample again it will be a different value, right because I updated my θ right, so this will essentially this will be θ old right.

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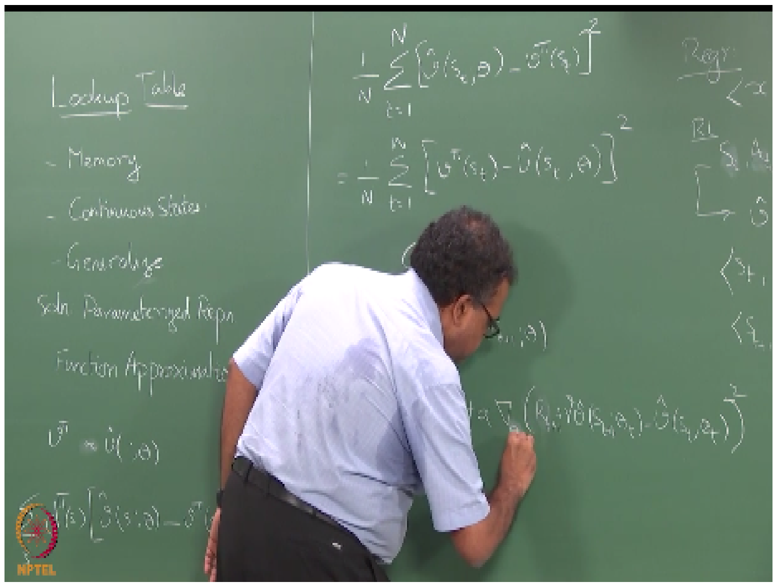

So I will be using the old θ, i will make a prediction i will make a target right, for my s1, so v s1 v st should become this where this value was computed using the old θ now I updated the θ, right. Once I do this, I update the θ, I go back and look at the target for St it will be R t + 1 + γ vhat s t + 1 θ-new right, it will be a different target, right so I am essentially trying to shoot a moving target right, so my targets keep changing after every iteration of updation so it is not like I can actually converge to that right, so lot of people thought this is all hukum, right I mean how can I use RL to train a neural network because I am shooting at a moving target right.

So people said RL would not work blah, blah and things like but then of course we always have this existential proof so luckily Gerry Tesauro never heard of all of these convergence results and stuff like that and so he fired up a neural network that learned to play backgammon, not a simple single neuron or anything it was a very complex neural network so he trained a neural network to play backgammon so people know that it works, right but seems to be pretty crazy that it actually works and later on there are lots of results that show that with linear parameterization, right.

We can show convergence even with this kind of an update one condition you need is that it should be what is the important condition I said on policy, sampling on trajectory sampling on policy so as long as you are being on policy you can converge that was shown right, so now we even have results for off policy updation right very recent things people have shown some kind of stability for off policy updates as well right. So, lot of work is being done in this space okay, because this is such a counterintuitive thing for people doing regression, right.

So every time I look at the data right, as soon as I use the data once the data is going to change right, so I cannot say that I have a stable convergent point but people have shown that this actually happens because of the way the dynamics of the system is set up, in fact the way you show that is that with the function approximation also the TD updates give you a contraction, right so we showed contraction long time back right so you have to show that eventually this even with the approximation it is a contraction under certain conditions on the approximation.

so that is essentially how people show this also works right, so any questions so far on this, okay. So how many of you are familiar with using gradient ascent or gradient descent to learn to optimize functions I expect every hand to go up because we talked about in policy gradient yes, can I assume that everybody knows this stuff right, otherwise flip back to reinforce so we talked about all of this in the policy gradient thing, right.

So in the policy gradient we did ascent right, why did we do ascent gradient ascent because the function we were dealing with there was the performance right and we want to maximize the performance with respect to the parameters θ, right. But the function I am dealing with here is the error I want to minimize the error with respect to the parameters θ so i will go in the opposite direction of the gradient right.

In policy gradient we went in the direction of the gradient here I will go in the opposite direction of the gradient till I try to get to the lowest point one of the popular ways of trying to minimize mean square error right, and a very popular variant of gradient descent is stochastic gradient descent so that I do this on a point wise basis so as soon as I get a single point estimate I go in the opposite direction of the gradient I estimate there, right.

So essentially what I can do now is have a truly online TD with function approximation implementation so every time I go to a state right every time I do not do the summation business here so every time I go to a state I form this estimate right give it to my SGD optimizer, it takes a step, right now I have a new function so now if I am going to do that at every time step I can start calling this as θ t, right.

So essentially what I am going to start doing now is I am going to say my θ t + 1 will be θ t + some α times gradient of the error right, where the error is computed as whatever so what you want to use, right so this is gradient of the error function evaluated at θ t, makes sense this will be my update rule, so one thing to notice there is that all the computation there is not $\hat{v}$ S t+1 θ t+1, it is S t+1 θ t, so depending on the actual form we choose for $\hat{v}$ right, so you will have different ways of computing this gradient and you can compute this once you compute the gradient you can actually make the update, so any questions on this so far.

How many of you are following it how many of you are completely lost anyone completely lost is that, any questions, good. So let us move on so there are many ways in which you can choose this parameterization right so the θ.

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